WP1: CDO & Arrival Sequencing at Barcelona – El Prat Airport (LEBL)

Infraestructures del Transport Aeri, 2023 – 24

Joel Rojas Llauradó
EETAC, Universitat Politècnica de Catalunya
Castelldefels, Spain
joel.rojas@estudiantat.upc.edu

Sara Allende Castilla
EETAC, Universitat Politècnica de Catalunya
Castelldefels, Spain
sara.allende@estudiantat.upc.edu

Jairo López Carbó
EETAC, Universitat Politècnica de Catalunya
Castelldefels, Spain
jairo.lopez@estudiantat.upc.edu

Àlex Fernández Ortiz
EETAC, Universitat Politècnica de Catalunya
Castelldefels, Spain
alex.fernandez.ortiz@estudiantat.upc.edu

***Abstract*—This project focuses on understanding Continuous Descent Operations (CDOs) and evaluating their potential implementation in arrivals at Barcelona – El Prat Airport (LEBL). Furthermore, a program has been developed to simulate various aircraft trajectories and analyse two CDO-based arrival scenarios. Finally, the feasibility of introducing CDOs at Barcelona Airport while ensuring safety and efficiency in aircraft altitudes will be assessed along Standard Terminal Arrival Routes (STARs).**

Keywords-component: Continuous Descent Operations (CDOs), Barcelona – El Prat Airport (LEBL), Standard Terminal Arrival Route (STAR).

I. INTRODUCTION

In recent years, the aviation industry has witnessed a significant surge in air travel, resulting in congestion on some airways and airports. To optimize aircraft management, techniques like Continuous Descent Operations (CDOs) have been developed and are the focal point of this project.

The primary objective of this task is to assess the impact of implementing Continuous Descent Operations (CDOs) in arrivals at Barcelona – El Prat Airport (LEBL). The assessment will weigh the advantages and disadvantages of CDOs, among many other important aspects, to determine their viability. Moreover, two scenarios involving CDO-based operations within various Standard Terminal Arrival Routes (STARs) procedures will be examined.

The project methodology encompasses researching key concepts such as CDOs and related technology. A simulation program capable of generating different aircraft trajectories has been created, accounting for all relevant parameters that influence these trajectories. Lastly, this simulator is applied to two different scenarios to analyse and derive conclusions.

II. FUNDAMENTAL ASPECTS OF CDOS

A. Background

A Continuous Descent Operation (CDO) is an aviation maneuver which facilitates the descent of arriving aircraft, following a continuous and steady flight path. It yields significant environmental and economic benefits and, for this reason, CDOs play an important role in creating a more efficient and sustainable air transportation.

With these operations, aircraft use reduced engine thrust and a low drag configuration. Critical factors include coordination between airspace and procedure design and Air Traffic Control (ATC).

B. Advantages

Continuous Descent Operations (CDOs) offer several key advantages. Firstly, they lead to significant fuel savings as aircraft follow optimal flight paths, reducing intermediate level-offs and, consequently, lowering fuel costs. According to Eurocontrol, CDOs save around 340000 tons of fuel annually in Europe [1].

Secondly, CDOs result in reduced emissions and noise pollution compared to traditional stepwise descents. This contributes to environmental sustainability by preventing the release of over 1 million tons of CO_2 and decreasing noise levels by 1 to 5 decibels [1].

Additionally, CDOs make it easier for Air Traffic Controllers (ATCs) and aviation companies to predict arrival times, enhancing schedule accuracy and flight punctuality.

Lastly, CDOs provide smoother descents, reducing the number of altitude and speed changes, which minimizes turbulence encounters and collisions, resulting in a more comfortable and pleasant flying experience for passengers.

C. Drawbacks

Continuous Descent Operations (CDOs), do come with some drawbacks. Firstly, they pose challenges for Air Traffic Controllers (ATCs) in predicting aircraft trajectories, often requiring reduced flight volumes or usage during off-peak traffic times, which impacts airport efficiency.

Secondly, CDOs are weather-dependent and can be disrupted by adverse conditions, such as turbulence, thunderstorms, or strong winds, adding complexity for pilots and ATCs.

Despite these limitations, the drive to reduce fuel consumption, emissions, and noise pollution remains a significant motivation for the aviation industry to improve the effectiveness of CDOs.

D. Technology required for its implementation

Continuous Descent Operations (CDOs) in aviation rely on key technologies and systems:

- Navigation Satellite Systems like GPS provide real-time aircraft position data for precise CDO route following.
- Flight Management Systems (FMS) aid in planning and executing optimal flight paths, including CDO profiles.
- Automatic Flight Control Systems (AFCS) enable automated control for maintaining the descent profile.

Pilots receive specialized training for these systems and effective real-time communication is crucial to perform safe CDOs.

E. Current implementation status

In 2021, Spanish airports, under ENAIRE, excelled in implementing Continuous Descent Approaches (CDAs), with 36% of authorized approaches, surpassing the European average of 18% [2].

Málaga-Costa del Sol airport led major European airports with 47% CDAs, while Barcelona-El Prat had 36% [2].

This initiative is in line with ENAIRE's 2025 Flight Plan, prioritizing environmental sustainability by reducing emissions and fuel consumption through CDAs, reaffirming their commitment to a greener aviation future.

III. SIMULATION

A simulation has been created to represent the trajectories followed by 5 different aircraft (B767-300ER, B777-300, B737, A320-212, A319-131) descending in two Maximum Landing Weight (MLW) configurations.

A. Data source

The program has been designed to accurately represent the aircraft trajectories in accordance with the specific performance models and data outlined in the project's Statement of Work (SoW). This customization is necessary because each aircraft exhibits its own unique performance characteristics.

It is expected that all aircraft will adhere to the practice of minimizing the rate of descent (ROD) an operating at idle thrust until reaching the Initial Approach Fix (IAF), at an altitude of 5000 feet. Furthermore, it is essential to ensure that the weight of the aircraft at the IAF does not exceed its Maximum Landing Weight (MLW).

The atmospheric conditions will conform to the International Standard Atmosphere (ISA).

B. Simulator development overview

The simulator is built using the Python programming language, which offers the advantage of allowing the creation of various functions with ease, without delving into too many technical intricacies. This enables the user to obtain the necessary parameters based on altitude for achieving a continuous descent profile for each aircraft.

The process begins with the getCDO(aircraft_model, MLW_percent) function, which takes two parameters: the aircraft model for which we want to determine the descent profile and the percentage of MLW at which it intercepts the Initial Approach Fix (IAF).

Initially, the code takes the string parameter referring to the desired aircraft model and searches within an aircraft dictionary to retrieve the necessary parameters for the calculations. It collects the relevant values and creates a new object of the aircraft class, which contains the parameters extracted from the dictionary and, therefore, makes calculations more straightforward.

Subsequently, the code enters in a loop with the aim of constructing two vectors: one for positions along the X-axis and another for positions along the Y-axis. By using a delta_t of -1, it is able to recreate the optimal continuous descent trajectory.

To determine the pair of values that dictate the aircraft's position at each moment, the equations mentioned in the following section are employed. Thus, each time the code enters the loop, all the values associated with Pressure, Temperature, Air Density, Thrust, Drag, Velocity, Angle, Mass, and Fuel Flow are recalculated. All prior positions are obtained using the Euler's method.

Once all the previous positions until reaching an altitude of 12000 meters have been calculated, the values in a two-dimensional graph can finally be represented. The X-axis represents the distance of the aircraft to the IAF and, the Y-axis, the altitude above sea level.

C. Equations

The following equations are used in the code development process:

To begin with, the force equations of a descending aircraft are formulated.

$$(1)\quad T - D - m \cdot g \cdot sin\gamma = 0$$

$$(2)\quad L - m \cdot g \cdot cos\gamma = 0 \rightarrow L = m \cdot g\ (cos\gamma = 1)$$

$T = thrust\ [N]$
$D = drag\ [N]$
$m = mass\ [kg]$
$g = gravity\ [m/s^2]$
$\gamma = angle\ of\ descent\ [rad]$
$L = lift\ [N]$

The expressions to obtain pressure, temperature and air density following the International Standard Atmosphere (ISA) [9] are the following.

<u>Temperature:</u>
Within the troposphere (up to 11000 m):

$$(3)\quad T = T_o - 6{,}5 \cdot {}^{h}/_{1000}$$

Within the tropopause (from 11000 m up to 19995 m):

$$(4)\quad T = constant = 216{,}65\ K\ (-56{,}5\ ºC)$$

$T = temperature\ [K]$
$T_o = temperature\ at\ mean\ sea\ level = 288{,}15\ K\ (15\ ºC)$
$h = altitude\ [m]$

<u>Pressure:</u>
Within the troposphere (up to 11000 m):

$$(5)\quad p = p_o \cdot \left(1 - 0{,}0065 \cdot \frac{h}{T_o}\right)^{5{,}2561}$$

$p = pressure\ [N/m^2]$
$p_o = pressure\ at\ mean\ sea\ level = 101325\ N/m^2\ (1013{,}25\ hPa)$

Within the tropopause (from 11000 m up to 19995 m):

$$(6)\quad p = p_{11} \cdot e^{-\frac{g}{R \cdot T_{11}} \cdot (h - h_{11})}$$

$p_{11} = pressure\ at\ the\ tropopause = 226{,}32\ hPa$
$T_{11} = temperature\ at\ the\ tropopause = 216{,}65\ K$
$h_{11} = altitude\ of\ the\ tropopause = 11000\ m$
$R = real\ gas\ constant\ for\ air = 287{,}04\ {}^{m^2}/_{K \cdot s^2}$

<u>Air density:</u>

$$(7)\quad \rho = {}^{p}/_{R \cdot T}$$

$\rho = air\ density\ [kg/m^3]$

Descent thrust (idle thrust) is determined taking into consideration both the maximum thrust and the pressure altitude.

If $h_p > h_{p,desc}$:

$$(8)\quad T_{desc} = CT_{Desc,high} \cdot T_{\max}$$

Else:
Clean configuration: (9) $T_{desc} = CT_{Desc,low} \cdot T_{\max}$
Approach configuration: (10) $T_{desc} = CT_{Desc,app} \cdot T_{\max}$

$T_{desc} = descent\ thrust\ [N]$
$T_{max} = maximum\ thrust\ [N]$

$CT_{Desc,high}$, $CT_{Desc,low}$ and $CT_{Desc,app}$ are thrust performance parameters specified in Eurocontrol's base of aircraft data (BADA) [8].

To continue, the expression to compute the maximum thrust.

$$(11)\quad T_{max} = C_{T1} \cdot \left(1 - \frac{h_p}{C_{T2}} + C_{T3} \cdot {h_p}^2\right)$$

$h_p = pressure\ altitude\ [ft]$

C_{T1}, C_{T2} and C_{T3} are aircraft dependent performance parameters specified in BADA [8].

Aircraft enter the approach configuration at an altitude of h_p = 6000 ft, and this transition leads to adjustments in the values of C_{D0} and C_{D2} due to modifications in the wing's aerodynamic surfaces.

Moreover, the expressions of drag, lift and drag coefficient are used.

$$(12)\quad D = \frac{1}{2} \cdot \rho \cdot S \cdot v^2 \cdot C_D$$

$S = wing\ surface\ [m^2]$
$v = velocity\ [m/s]$
$C_D = drag\ coefficient$

$$(13)\quad L = \frac{1}{2} \cdot \rho \cdot S \cdot v^2 \cdot C_L$$

$C_L = lift\ coefficient$

$$(14)\quad C_D = C_{D0} + C_{D2} \cdot {C_L}^2$$

$C_{D0} = parasite\ drag\ coefficient$
$C_{D2} = induced\ drag\ coefficient$

C_{D0} and C_{D2} are two parameters specified in BADA [8] as a function of the aircraft and its flaps/slats configuration (approach or clean).

After that, the minimum Rate of Descent (ROD) velocity is gotten taking into consideration the ROD expression.

$$(15)\quad ROD = v \cdot sin\gamma$$

$ROD = Rate\ of\ Descent\ [m/s]$

Isolating the velocity, we finally obtain the expression for the speed corresponding to the minimum Rate of Descent.

$$(16)\quad v_{\min ROD} = \sqrt{\frac{T \pm \sqrt{T^2 + 12 \cdot m^2 \cdot g^2 \cdot C_{D0} \cdot C_{D2}}}{3 \cdot \rho \cdot S \cdot C_{D0}}}$$

In addition, note that the angle of descent is obtained isolating this parameter from the force equations.

$$(17)\quad \gamma = arcsin\left({}^{T - D}/_{m \cdot g}\right)$$

Next, the equation to compute the thrust specific fuel consumption.

$$(18)\quad \eta = C_{F1} \cdot \left(1 + \frac{v}{C_{F2}}\right)$$

$\eta = thrust\ specific\ fuel\ consumption$ [kg/ min· kN]
$v = true\ airspeed\ [kt]$

C_{F1} and C_{F2} are aircraft dependent parameters specified in BADA [8].

The fuel flow is gotten with the following expression.

$$(19)\quad FF = \eta \cdot T$$

$FF = fuel\ flow$ [kg/min]
$T = thrust\ [N]$

Finally, the Euler's method is used to obtain positions.

$$(20)\quad X_n = X_{n-1} + \Delta X \cdot delta_t \rightarrow \Delta X = cos\gamma \cdot v$$
$$(21)\quad Y_n = Y_{n-1} + \Delta Y \cdot delta_t \rightarrow \Delta Y = cos\gamma \cdot v$$

$X_n, Y_n = current\ position$
$X_{n-1}, Y_{n-1} = prior\ position$
$delta_t = -1$

D. Results

The 10 trajectories plotted by the simulator are represented in Figure 1 in the 'Appendix'.

From the plot, the altitude at which an aircraft is flying at a certain distance from the Initial Approach Fix (IAF), located at x = 0, can be known.

IV. Sequencing Analysis

An analysis has been conducted regarding a multiple CDO-based arrival scenario at RWY 24R in LEBL.

It has been considered that the altitude of all aircraft at the Standard Terminal Arrival Route (STAR) entry point is the one required to intercept the Initial Approach Fix (IAF) at 5000 ft.

The Standard Terminal Arrival Routes' (STARs) map in Barcelona Airport is represented in Figure 2 in the 'Appendix'.

A. Scenario 1

Scenario 1 consists on having simultaneous arrivals at the STAR entry points.

In addition, it must be taken into account that the calculation of the arrival times for various aircraft at the Initial Approach Fix (IAF) consider that all aircraft reach the first waypoint (WP) of the Standard Terminal Arrival Route (STAR) at precisely 10:25:00 AM.

The results table is represented in Figure 3 in the 'Appendix', along with a plot showing the STAR entry point within the trajectory of every aircraft (Figure 4).

Considering that both maximization of runway throughput and maintenance of safety levels are required, it has been considered that, even though this scenario provides efficiency in relation to the runway utilization, due to the fact that, on average, the planes are 101, 8 seconds apart when they cross the Initial Approach Fix (IAF), it is not feasible since the security distance is not respected at all times.

The most notable case is that of the fifth aircraft entering the approach segment (B737), which is only 30 seconds behind the preceding plane (A320-212). Taking into account that both aircraft belong to the Medium (M) category and that the approach speed of the succeeding aircraft (at MLW) is between 121 and 141 kt [10], the minimum security distance of 4 NM is not respected.

A potential pre-tactical action to achieve a more optimum scenario could consist on separating both aircraft enough before entering the approach segment to maintain a minimum level of safety once they are in it. In doing so, the scenario could be applied and benefits regarding the maximization of runway throughput would surge, because the planes would land relatively close maintaining safety.

B. *Scenario 2*

Scenario 2 consists on optimizing arrivals sequencing at the Initial Approach Fix (IAF).

To enhance runway throughput while upholding safety standards, the aim of approach Air Traffic Control Officers (ATCOs) is to establish a consistent rate of aircraft arrivals at the Initial Approach Fix (IAF). In pursuit of this goal, it is essential to acknowledge that approach controllers can collaborate with en-route controllers to determine the arrival times of each aircraft at the Standard Terminal Arrival Route (STAR) entry point.

New times of arrival at the STAR entry point have been proposed taking into consideration the following aspects: the time separation between aircraft at IAF is 2 minutes, the time of arrival at the STAR entry point can be advanced or delayed with respect to the time defined in the first scenario and no time changes will be applied to the first aircraft arriving to the IAF. Furthermore, among the multiple possible combinations, the one maintaining the same order of arrival at the IAF has been selected.

The results table is represented in Figure 5 in the 'Appendix'.

Finally, when it comes to the benefits of this scenario, it can be deduced that it is efficient in relation with the runway throughput because of the fulfillment of the required conditions. Nevertheless, to achieve it, it is necessary to advance the arrival at the STAR entry waypoint of a specific aircraft (A319-131) which can cause a redistribution of flights in the en-route phase and a subsequent extra work for the corresponding Air Traffic Controllers (ATCs).

C. *Scenarios' overview*

To evaluate the suitability of the simulated trajectories considering the current STAR altitude constraints at Barcelona – El Prat Airport (LEBL), a table has been developed (Figure 6 in the 'Appendix').

Analysing the results, it has been concluded that the simulated trajectories do not fit the STAR altitude constraints and, therefore, are not suitable.

CONCLUSIONS

Once all the study on Continuous Descent Operations (CDOs) has been done, it is obvious that these types of operations offer numerous advantages, such as a decrease of fuel consumption and pollution.

Despite all these benefits, after comparing the results obtained in the analysis, in both Scenario 1 and Scenario 2, with the approach chart of RWY 24R in Barcelona – El Prat Airport (LEBL), it can be concluded that it is not possible to implement CDOs in LEBL. The main reason is because the altitude restrictions on the waypoints (WP) which compose the Standard Terminal Arrival Routes (STARs) of this runway make impossible implementing these operations due to the fact that in all the cases, and for all the Maximum Landing Weights (MLWs), this altitude is out of the limits.

In conclusion, nowadays and with the AIP that ENAIRE offers, it is not efficient to implement Continuous Descent Operations in Barcelona – El Prat Airport. Nevertheless, implementing CDOs in the aviation industry would be a significant advancement. An improvement on the techniques and operations used would help reach a revolutionary future in the air transport industry.

REFERENCES

[1] Eurocontrol, "Continuous climb and descent operations", (https://www.eurocontrol.int/concept/continuous-climb-and-descent-operations).

[2] ENAIRE,"ENAIRE improves flight efficiency by making greater use of continuous descent operations"(https://www.enaire.es/en_GB/2022_09_27/ndp_enaire_improves_continuous_descent_operations).

[3] R. Sáez, X. Prats, "Comparison of Fuel Consumption of Continuous Descent Operations with Required Times of Arrival", ICRAT 2020.

[4] J. Robinson, M. Kamgarpour, "Benefits of Continuous Descent Operations in High-Density Terminal Airspace Under Scheduling Constraints".

[5] D.Toratani, N. Wickramasinghe, J. Westphal, T. Reuerle, "Feasibility study on applying continuous descent operations in congested airspace with speed control functionality: Fixed flight-path angle descent", Aerospace science and technology, 2020, Vol.107, p.106236, Article 106236.

[6] D. Gui, M. Le, Z. Huang, J. Zhang, A. D'Ariano, "Optimal aircraft arrival scheduling with continuous descent operations in busy terminal maneuvering areas", Journal of Air Transport Management, Volume 107, March 2023, 102344.

[7] R. Sáez, T. Polishchuk, C. Schmidt, H. Hardell, L. Smetanová, V. Polishchuk, X. Prats, "Automated sequencing and merging with dynamic aircraft arrival routes and speed management for continuous descent operations", Transport Research Part C: Emerging Techologies, Volume 132, November 2021, 103402.

[8] Eurocontrol Experimental Centre, User Manual for the Base of Aircraft Data (BADA) Revision 3.10, EEC Technical/Scientific Report No. 12/04/10-45, April 2012.

[9] M. Cavcar, "The International Standard Atmosphere (ISA)", Anadolu University, Turkey.

[10] SKY brary, "Mitigation of wake turbulence hazard" (https://skybrary.aero/articles/mitigation-wake-turbulence-hazard).

APPENDIX

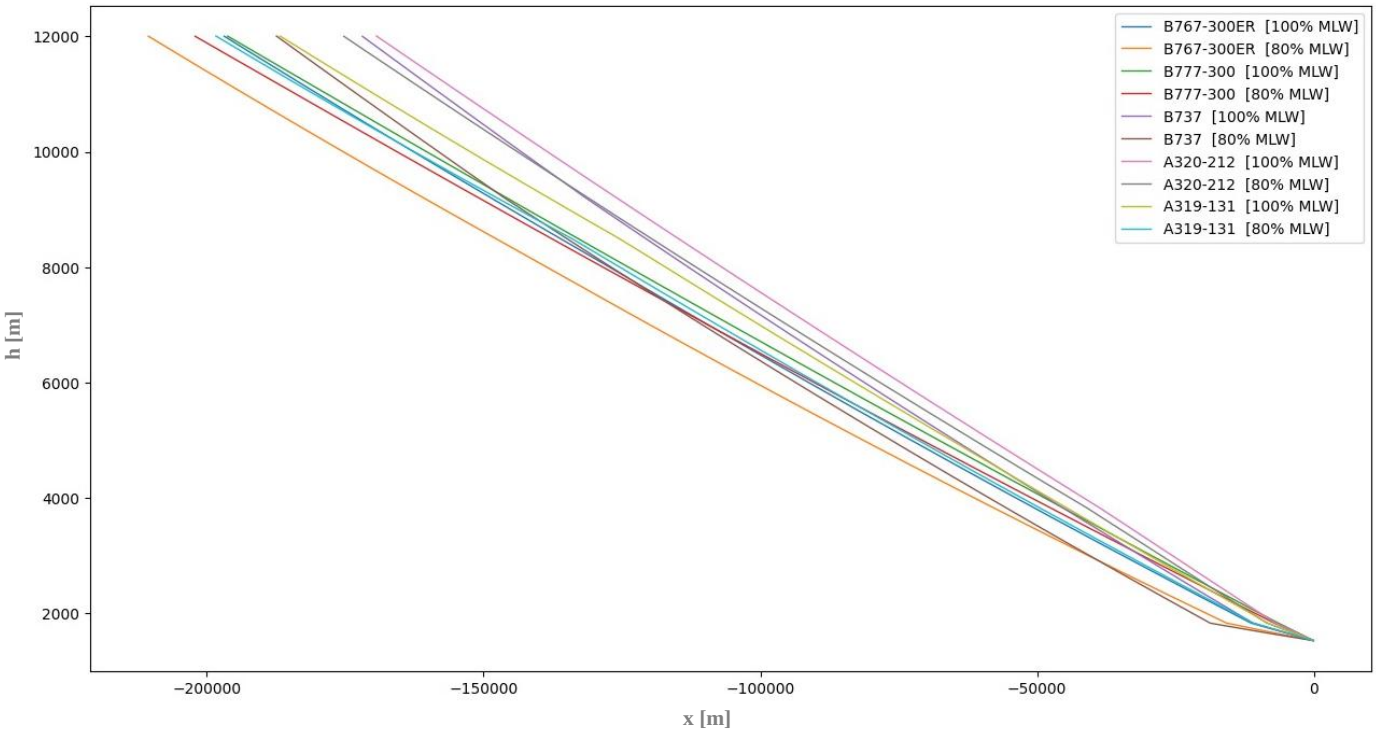

Figure 1. The 10 trajectories plotted by the simulator.

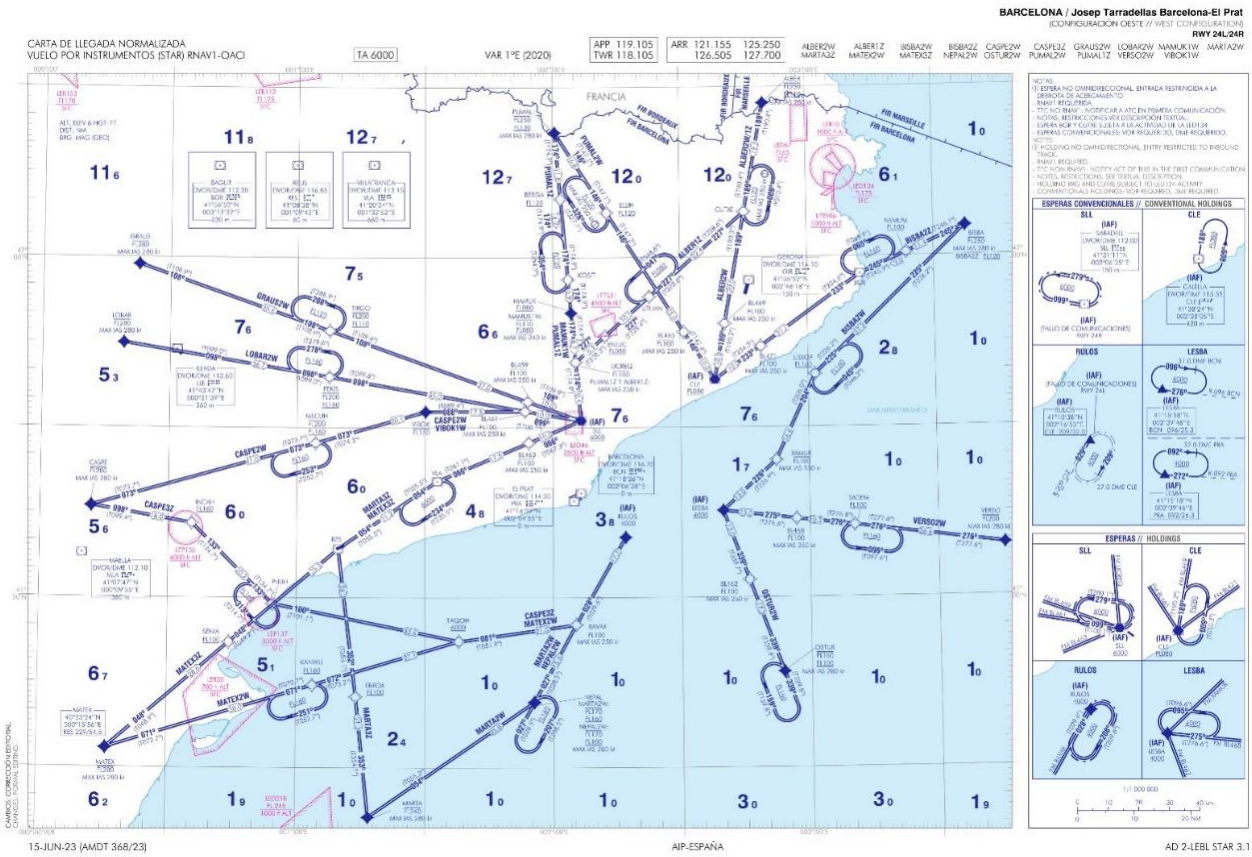

Figure 2. Standard Terminal Arrival Routes (STARs) used to analyse Scenarios 1 and 2.

Aircraft model	STAR	%MLW IAF	Dist. STAR entry WP to IAF [km]	Alt. at STAR entry WP [m]	Time from STAR entry WP to IAF [s]	ToA IAF	Seq. Order	Sep. to previous at IAF [s]
B767-300ER	MATEX2W	80	192.24	10950	1696	10:53:16	6	120
B777-300	MARTA2W	80	127.97	7982	1187	10:44:47	1	---
B737	CASPE2W	100	164.09	11457	1576	10:51:16	5	30
A320-212	GRAUS2W	100	152.98	10945	1393	10:48:13	3	66
A319-131	BISBA2W	100	123.90	8405	1327	10:47:07	2	140
A320-212	LOBAR2W	80	151.31	10470	1546	10:50:46	4	153

Figure 3. Results table of Scenario 1.

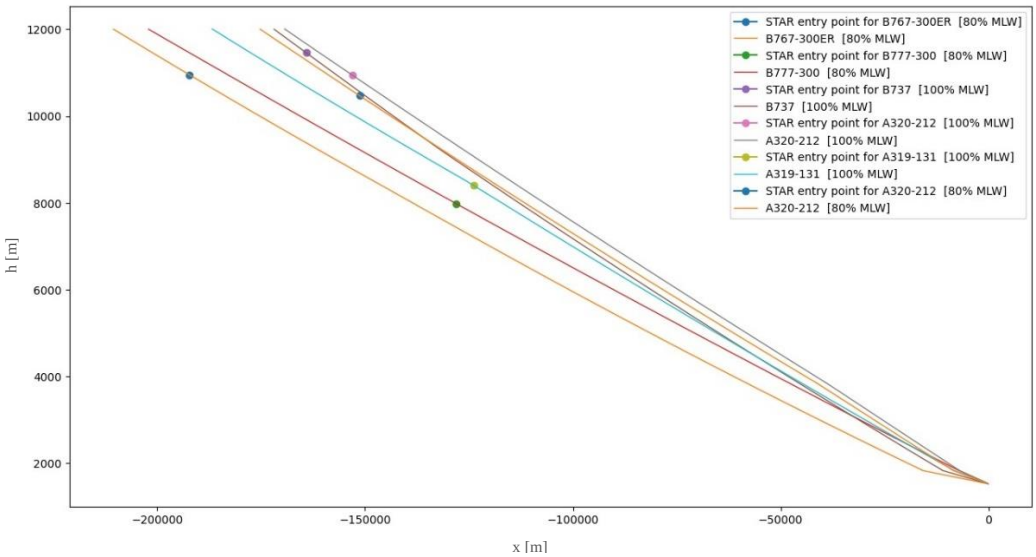

Figure 4. The STAR entry point within the trajectory of every aircraft plotted by the simulator.

Aircraft model	STAR	Weight at IAF	ToA IAF Scenario2	Applied delay [s]	ToA STAR entry WP Scenario 2
B767-300ER	MATEX2W	80	10:54:47	90	10:26:30
B777-300	MARTA2W	80	10:44:47	0	10:25:00
B737	CASPE2W	100	10:52:47	92	10:26:32
A320-212	GRAUS2W	100	10:48:47	34	10:25:34
A319-131	BISBA2W	100	10:46:47	-20 *(anticipa entrada)*	10:24:40
A320-212	LOBAR2W	80	10:50:47	1	10:25:01

Figure 5. Results table of Scenario 2.

ALTITUDE CONSTRAINS ANALYSIS TABLE

			WP1						WP2						WP3						WP4					
Aircraft Model	%MLW	STAR Name	WP Name	Distance to IAF [NM]	Lower Limit	Upper Limit	Actual Altitude [ft]	Inside Limits?	WP Name	Distance to IAF [NM]	Lower Limit	Upper Limit	Actual Altitude [ft]	Inside Limits?	WP Name	Distance to IAF [NM]	Lower Limit	Upper Limit	Actual Altitude [ft]	Inside Limits?	WP Name	Distance to IAF [NM]	Lower Limit	Upper Limit	Actual Altitude [ft]	Inside Limits?
B767-300ER	100	MATEX2W	MATEX	103,8	---	FL280	38486	NO	KANWU	65,8	FL160	---	25252	YES	TAQOH	38,7	6000	---	16219	NO	RAVAX	17,7	---	FL100	9562	YES
	80						35925	NO					23318	YES					14809	NO					8610	YES
B777-300	100						38678	NO					25898	YES					17083	NO					10440	NO
	80						37616	NO					25126	YES					16564	NO					10136	NO
B737	100						44122	NO					28141	YES					17770	NO					10140	NO
	80						40509	NO					25243	YES					15527	NO					8470	YES
A320-212	100						44364	NO					29345	YES					19089	NO					11336	NO
	80						43047	NO					28313	YES					18406	NO					10902	NO
A319-131	100						40471	NO					27181	YES					17562	NO					10392	NO
	80						38245	NO					25584	YES					16438	NO					9678	YES
B767-300ER	100	MARTA2W	MARTA	69,1	---	FL240	26398	NO	NEPAL	33,3	FL160	FL170	14474	NO	RAVAX	17,7	---	FL100	9562	YES	---					
	80						24373	NO					13178	NO					8610	YES						
B777-300	100						26998	NO					15368	NO					10440	NO						
	80						26186	NO					14884	NO					10136	NO						
B737	100						29418	NO					15780	NO					10140	NO						
	80						26467	NO					13680	NO					8470	YES						
A320-212	100						30619	NO					17090	NO					11336	NO						
	80						29571	NO					16469	YES					10902	NO						
A319-131	100						28331	NO					15705	NO					10392	NO						
	80						26734	NO					14685	NO					9678	YES						
B767-300ER	100	CASPE2W	CASPE	88,6	---	FL280	33094	NO	MECUH	47,6	FL160	FL200	19414	YES	VIBOK	27,5	---	FL150	12638	YES	BL461	10	---	FL100	7184	YES
	80						30768	NO					17531	YES					11464	YES					6425	YES
B777-300	100						33516	NO					19946	YES					13528	YES					8037	YES
	80						32538	NO					19341	YES					13120	YES					7840	YES
B737	100						37589	NO					21106	NO					13666	YES					7417	YES
	80						34139	NO					18651	YES					11710	YES					5991	YES
A320-212	100						38259	NO					22407	NO					14965	YES					8387	YES
	80						37030	NO					21604	NO					14433	YES					8032	YES
A319-131	100						35022	NO					20674	NO					13717	YES					7827	YES
	80						33098	NO					19396	YES					12799	YES					7277	YES
B767-300ER	100	GRAUS2W	GRAUS	82,6	---	FL280	31017	NO	TIRGO	47	FL110	FL200	18943	YES	BL459	10	---	FL100	7184	YES	---					
	80						28796	NO					17349	YES					6425	YES						
B777-300	100						31489	NO					19766	YES					8037	YES						
	80						30560	NO					19147	YES					7840	YES						
B737	100						35030	NO					20890	NO					7417	YES						
	80						31668	NO					18440	YES					5991	YES						
A320-212	100						35909	NO					22199	NO					8387	YES						
	80						34718	NO					21385	NO					8032	YES						
A319-131	100						32932	NO					20465	NO					7827	YES						
	80						31144	NO					19189	YES					7277	YES						
B767-300ER	100	BISBA2W	BISBA	66,9	---	FL250	25631	NO	USSOF	29,6	FL160	---	13294	NO	XAMUR	13	---	FL100	8114	YES	---					
	80						23674	NO					12086	NO					7265	YES						
B777-300	100						26279	NO					14188	NO					8960	YES						
	80						25477	NO					13759	NO					8730	YES						
B737	100						28571	NO					14429	NO					8468	YES						
	80						25633	NO					12423	NO					6940	YES						
A320-212	100						29775	NO					15731	NO					9537	YES						
	80						28728	NO					15161	NO					9149	YES						
A319-131	100						27575	NO					14433	NO					8823	YES						
	80						25970	NO					13485	NO					8215	YES						
B767-300ER	100	LOBAR2W	LOBAR	81,7	---	FL280	30680	NO	PEKIS	45	FL160	FL200	18290	YES	BL461	10	---	FL100	7184	YES	---					
	80						28496	NO					16735	YES					6425	YES						
B777-300	100						31206	NO					19121	YES					8037	YES						
	80						30267	NO					18512	YES					7840	YES						
B737	100						34669	NO					20122	NO					7417	YES						
	80						31322	NO					17728	YES					5991	YES						
A320-212	100						35541	NO					21443	NO					8387	YES						
	80						34349	NO					20657	NO					8032	YES						
A319-131	100						32605	NO					19770	YES					7827	YES						
	80						30837	NO					18525	YES					7277	YES						

Figure 6. Altitude constrains analysis table.